[7, 3, 2, 5, | 6, 1, 10, 2]

$$\begin{array}{l} n \\ [\,2, 3, 5, 7\,] \\ \;\cancel{\uparrow}\;\cancel{\uparrow}\;\cancel{\uparrow}\;\cancel{\uparrow}\;\uparrow \end{array} \qquad \begin{array}{l} m \\ [\,1, 2, 6, 10\,] \\ \;\cancel{\uparrow}\;\cancel{\uparrow}\;\cancel{\uparrow}\;\cancel{\uparrow}\;\uparrow \end{array}$$

[1, 2, 2, 3, 5, 6, 7, 10]

Merging → TC: O(n + m)
AS: O(1)

K=4

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & \cancel{1} & \underline{5} & 7 & 15 \\ 1 & \cancel{1} & \cancel{2} & \underline{3} & 6 \\ 2 & \underline{2} & 3 & 10 & 20 \\ 3 & \cancel{1} & \underline{3} & 4 & 5 \end{array}$$

$$\text{ptrs} \rightarrow \begin{array}{cccc} 0 & 1 & 2 & 3 \\ [\,\cancel{0}\,1 & \cancel{0}\,\cancel{1} & 0 & \cancel{0}\,1\,] \\ & 2 & & \end{array}$$

[1, 1, 1, 2

TC → O($\boxed{K^2}$ × $\boxed{K}$)
= O(K³)

AS → O(K)

$$\begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & \cancel{1} & 5 & 7 & 15 \\ 1 & \cancel{1} & \cancel{2} & \underline{3} & 6 \\ 2 & \underline{2} & 3 & 10 & 20 \\ 3 & \cancel{1} & 3 & 4 & 5 \end{array}$$

ptrs: ~~1, 1, 2, 1~~

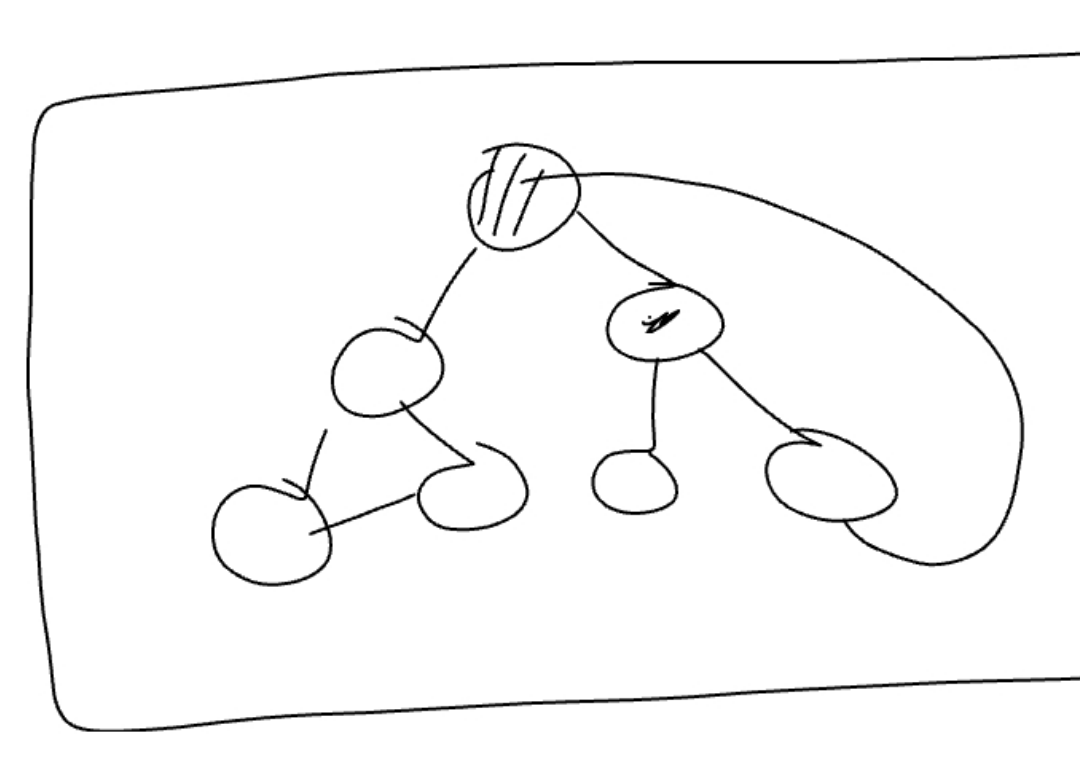
{~~1, (0,0)~~} {5, (0,1)}
{~~1, (1,0)~~} {~~2, (1,1)~~} {3, (1,2)}
{1, (2,0)}
{2, ~~(3,0)~~} {3, (3,1)}
{1, ~~(3,0)~~}

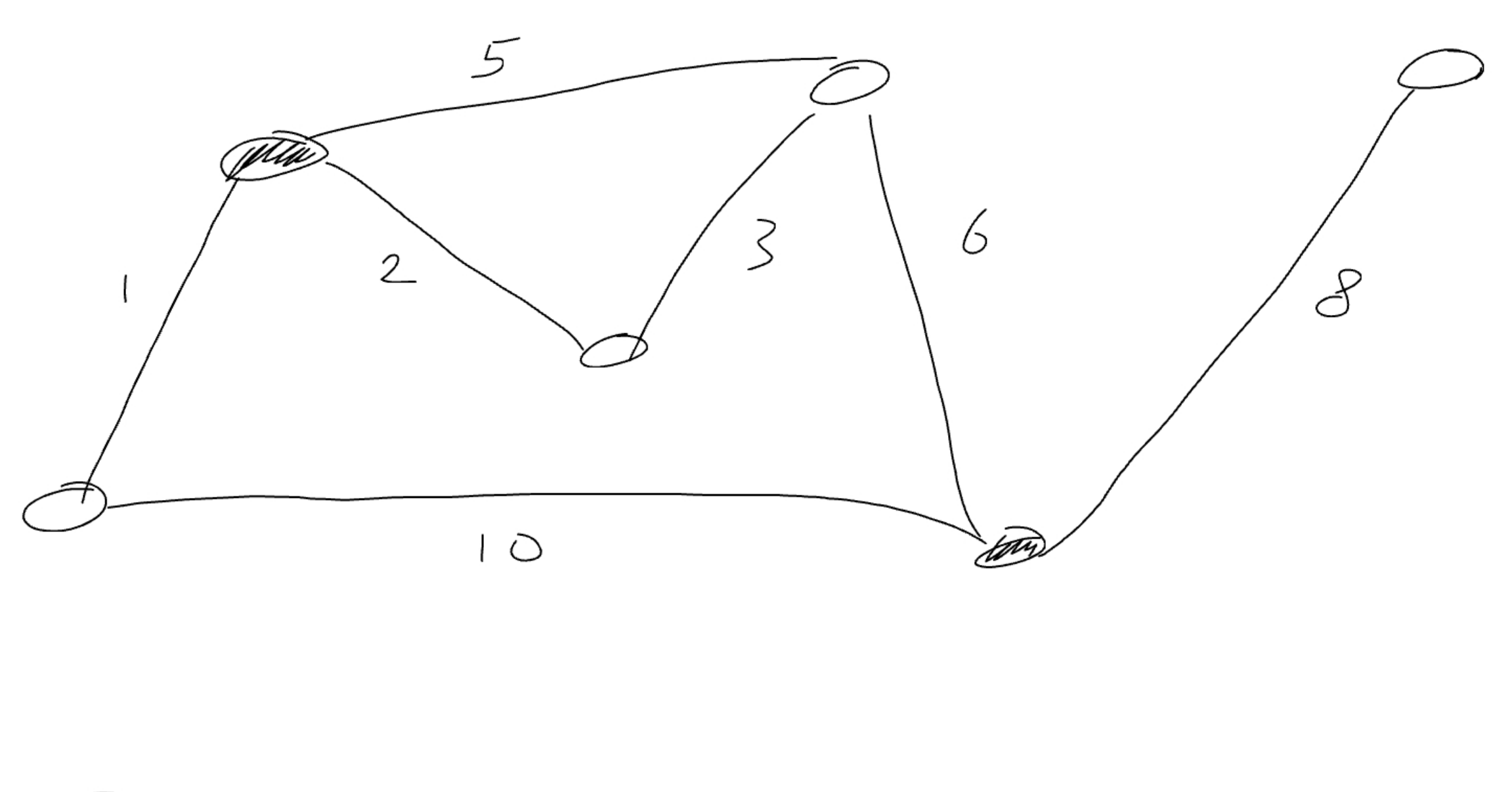
[1, 1, 1, 2 - - . . . .

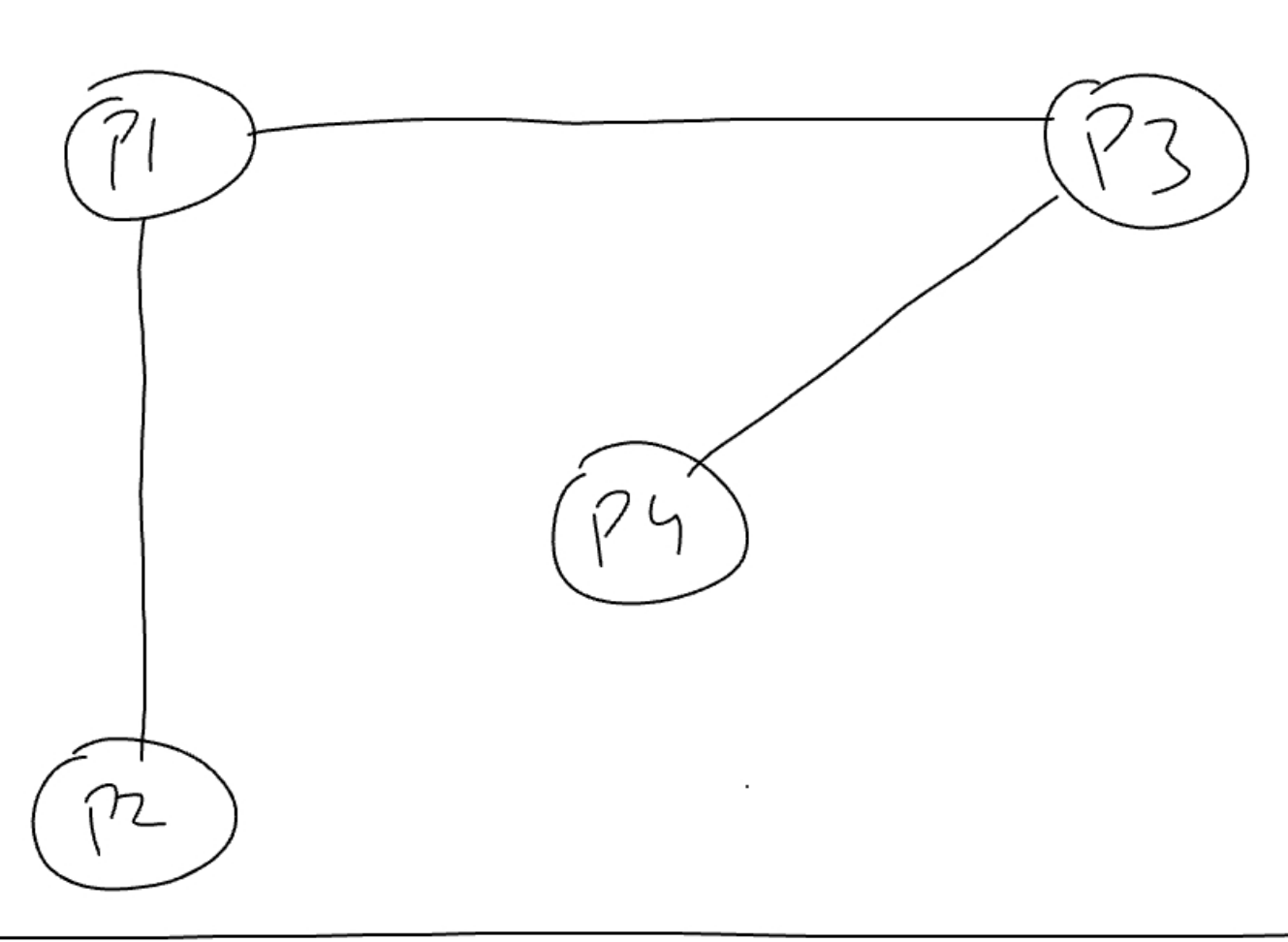
TC: O(K² × log(K))
AS: O(K)

(5, 3, 7, ②)

-5, -3, -7, -2

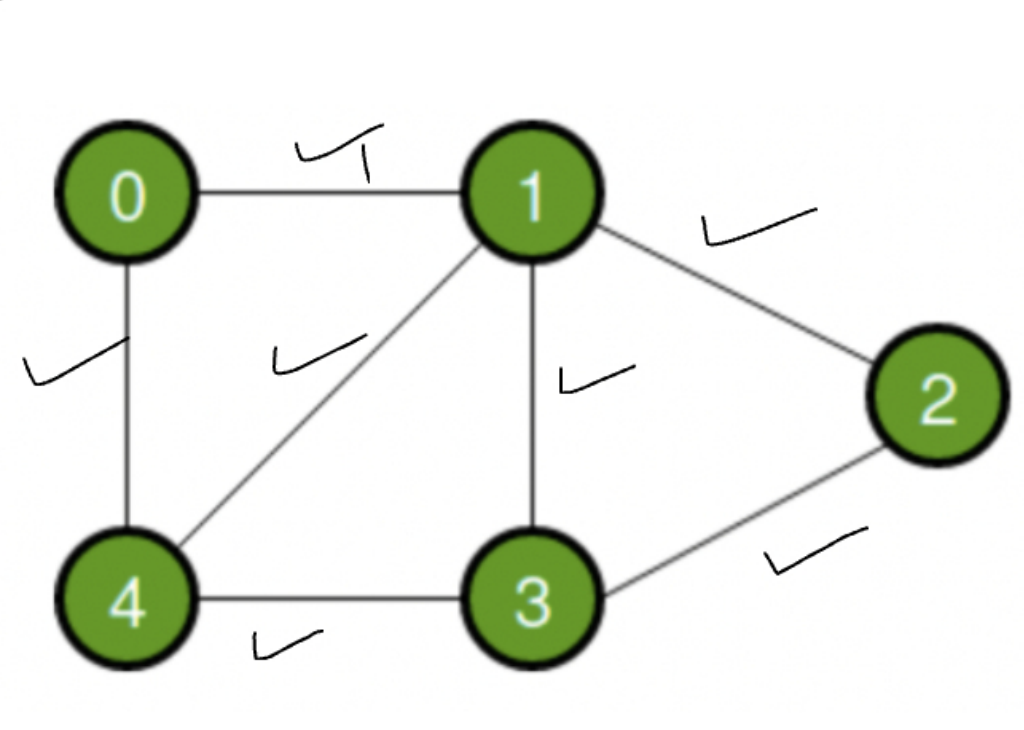






Adj Matrix

V = 5



space = O(V²)

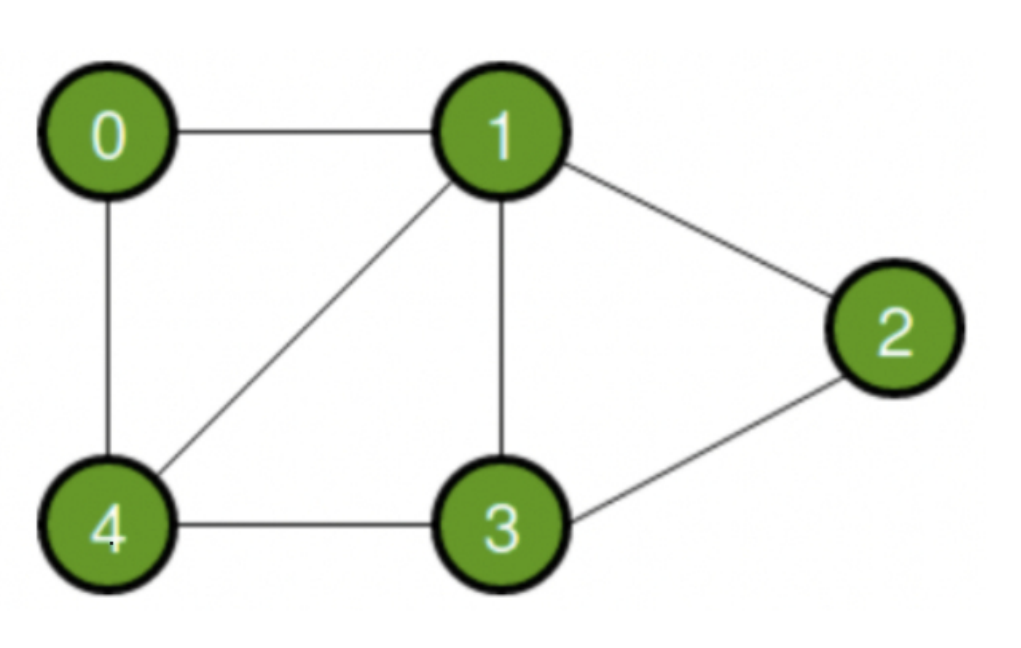
$$\begin{array}{c|ccccc} & 0 & 1 & 2 & 3 & 4 \\ \hline 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 & 1 & 0 \\ 3 & 0 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 0 & 1 & 0 \end{array}$$

adj[i][j] = 1 if i → j edge exists
= 0 otherwise

For weighted graphs:
adj[i][j] = w if $i \xrightarrow{w} j$ edge exists
= 0 otherwise

Adj List

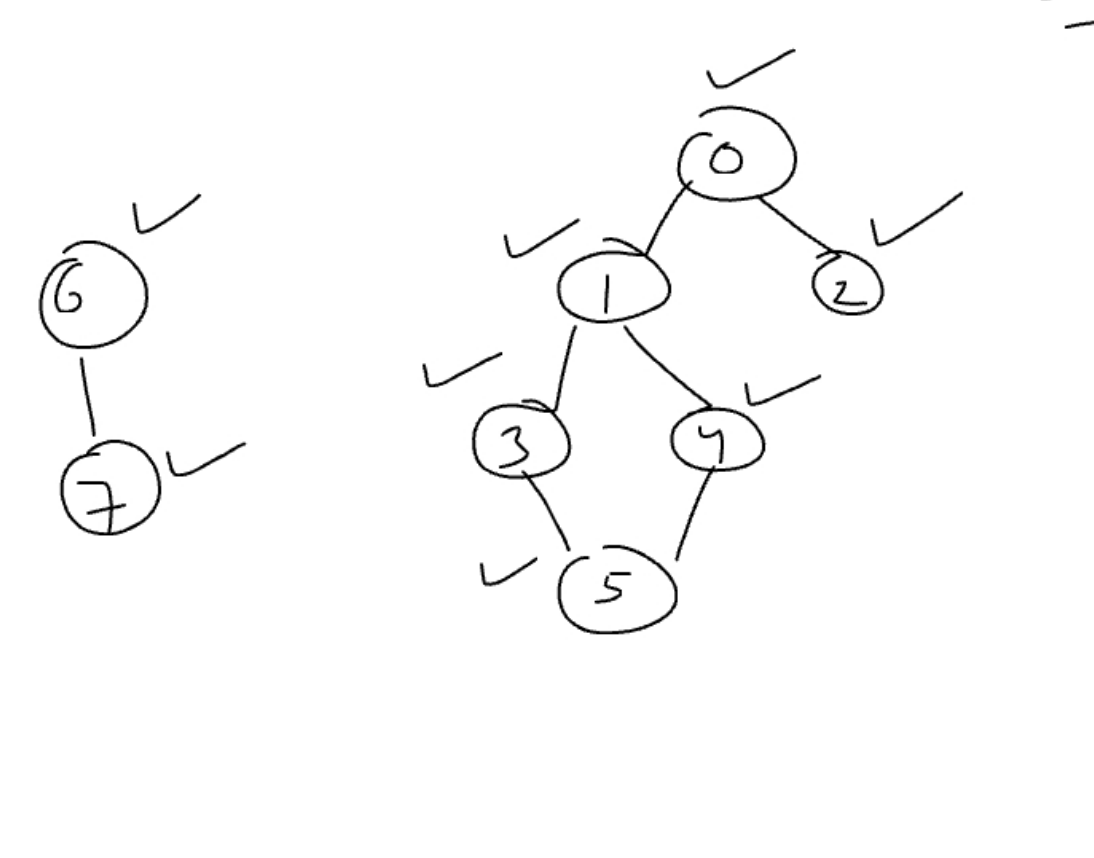
V, E



0 → [1, 4]
1 → [0, 4, 3, 2]
2 → [1, 3]
→ 3 → [2, 1, 4]
4 → [0, 1, 3]

space = O(V + E)

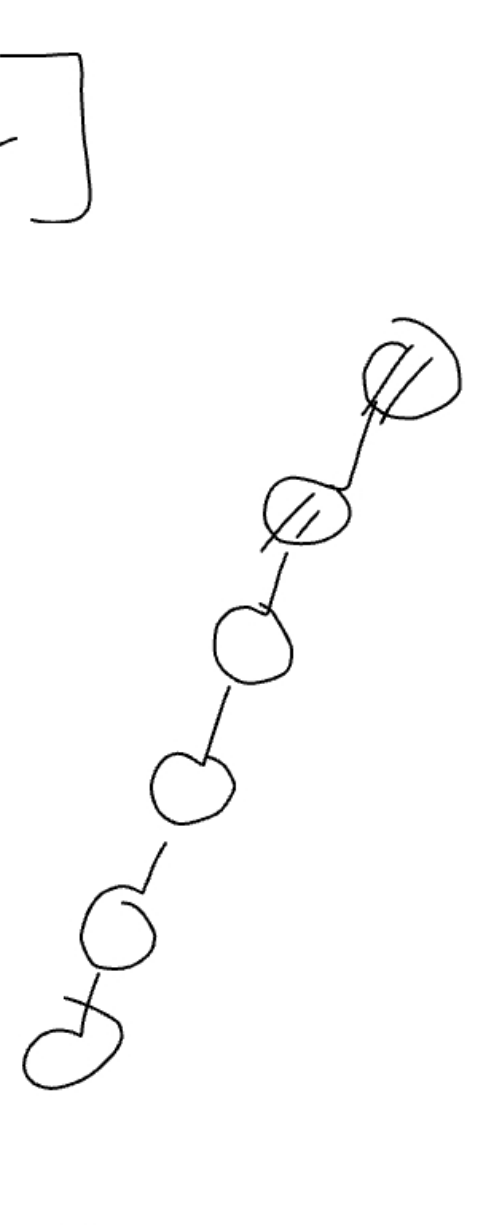
DFS

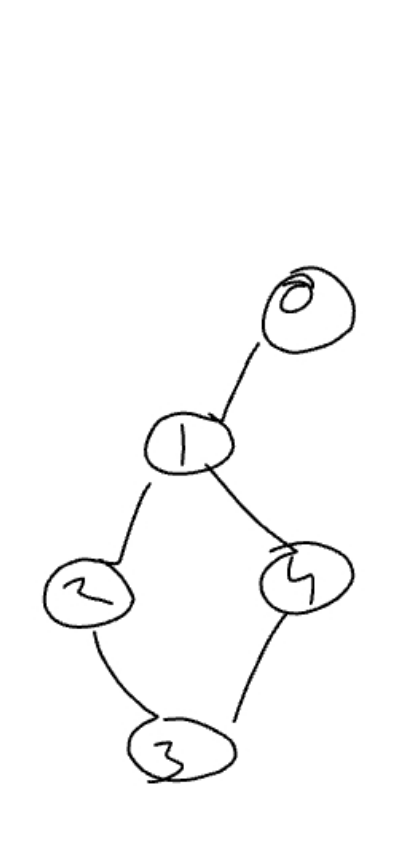


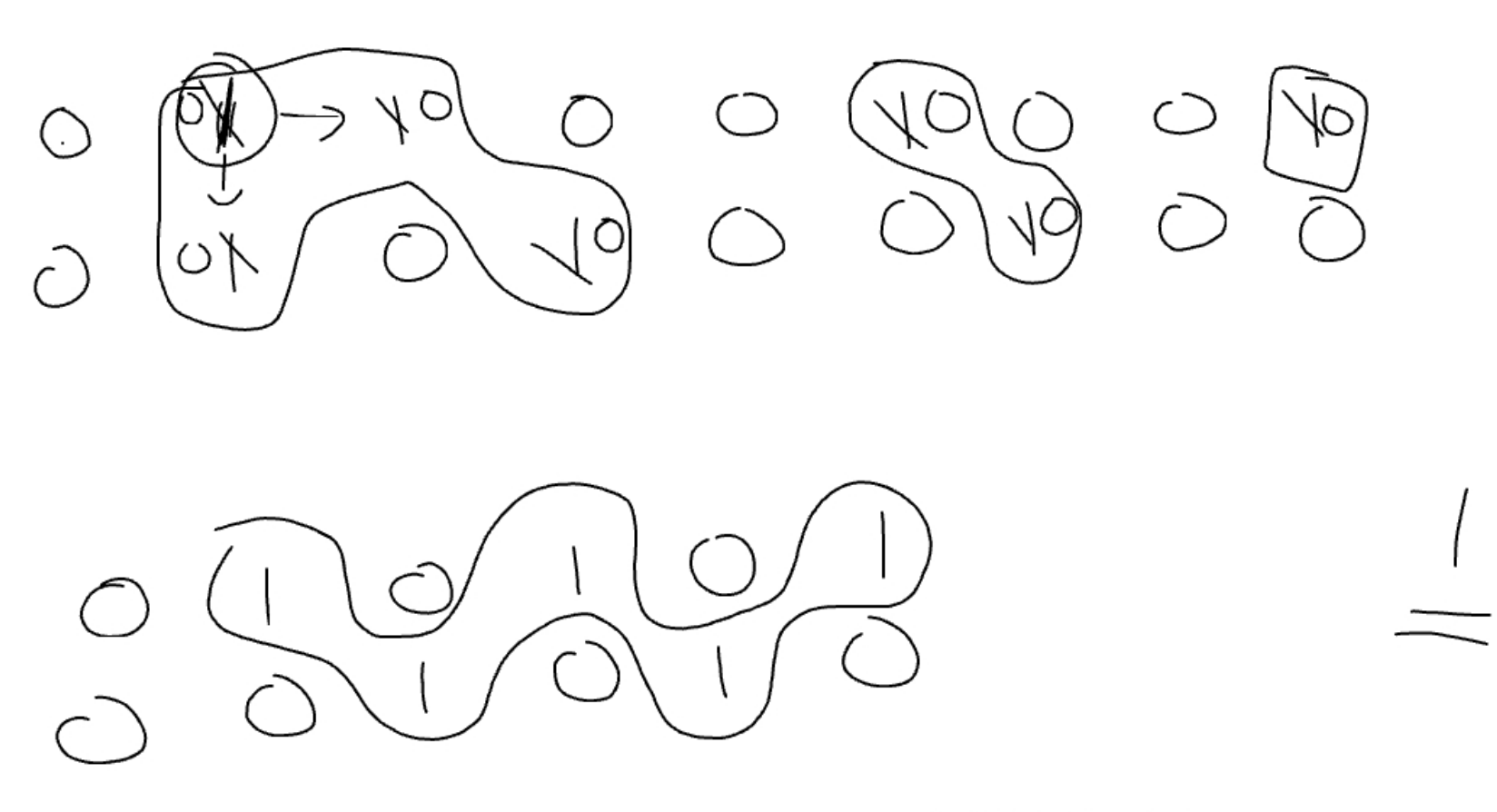
0 → (1, 2
1 → 0, 3, 4
2 → 0
3 → 1, 5
4 → 1, 5
5 → 3, 4
6 → 7
7 → 6)

~~0, 1, 3, 5, 2, 4, 3, 1~~

[0, 1, 3, 5, 4, 2]







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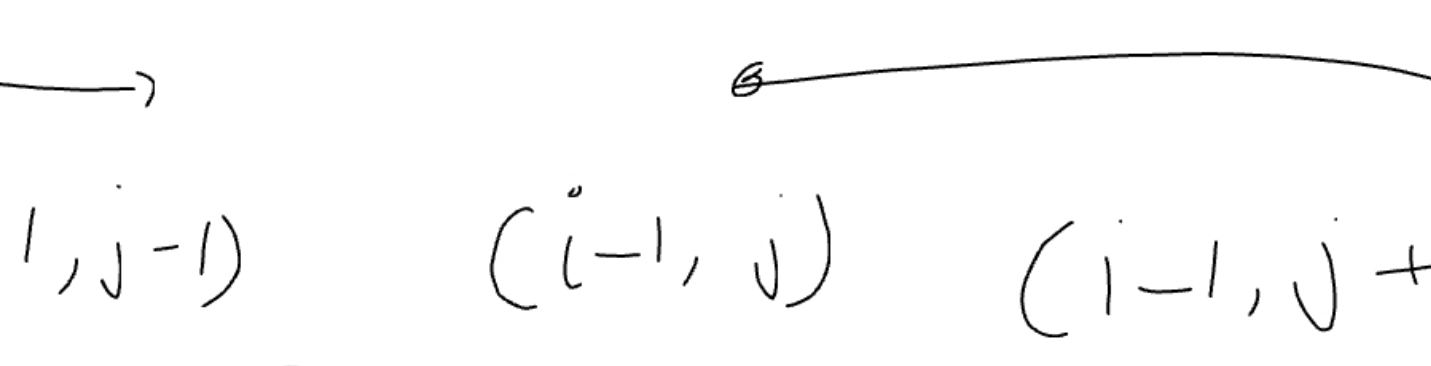

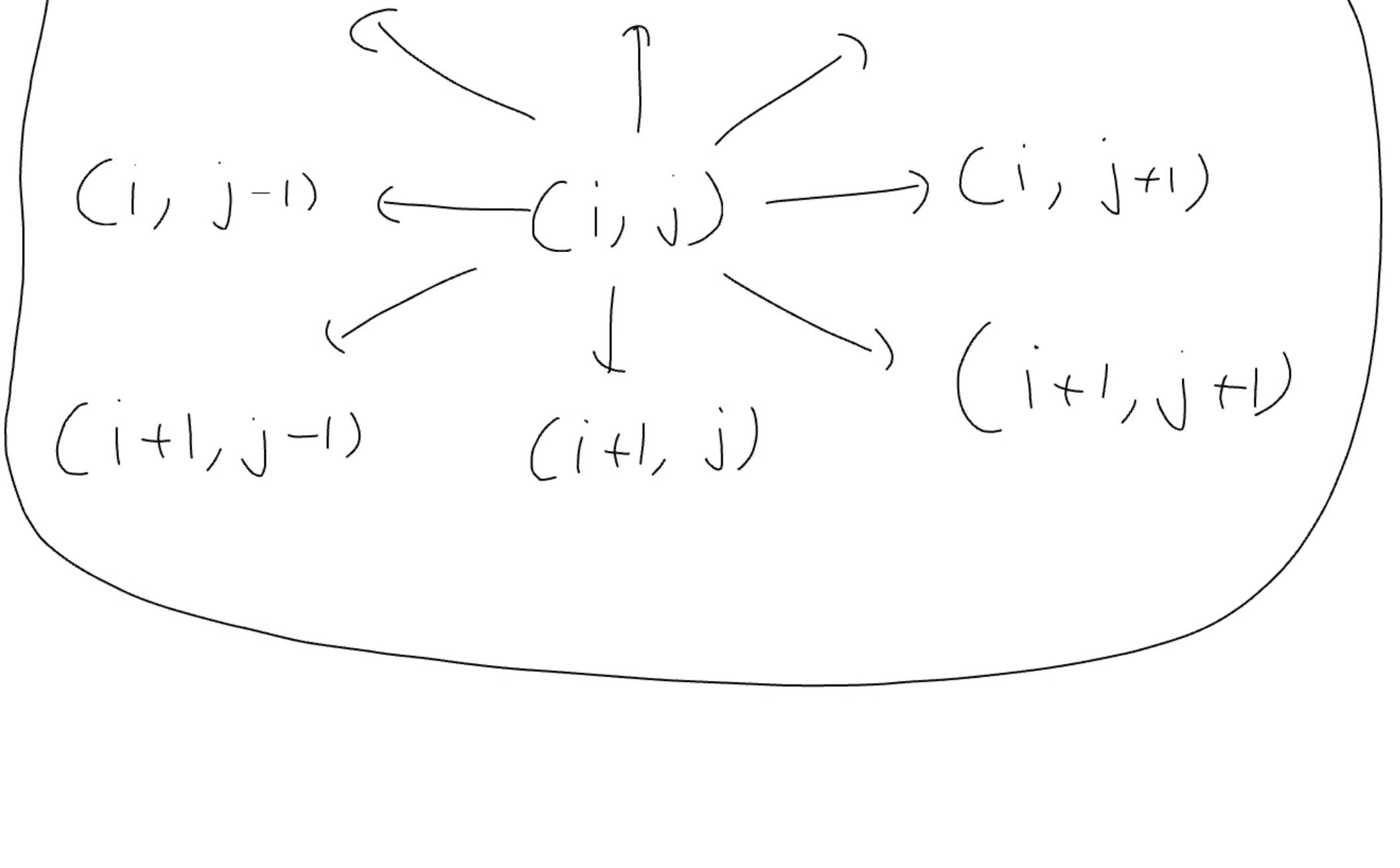1